

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA51 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5 TOTAL MARKS: 50

Annexure A

Comparative Analysis

Code of Conduct:

*I **T Sanhith(192311228)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A
Comparative Analysis

1. Compare Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPSec with respect to security services, performance overhead, and suitability for different network scenarios. Analyze which is more appropriate for a VPN deployment.

ANS: IPSec provides security for communication at the network layer. The two main IPSec protocols are **Authentication Header (AH)** and **Encapsulating Security Payload (ESP)**.

Comparison between AH and ESP

Feature	Authentication Header (AH)	Encapsulating Security Payload (ESP)
Confidentiality	Not provided	Provided through encryption
Integrity	Provided	Can be provided
Authentication	Provided	Can be provided
Encryption	No	Yes
Protection against packet modification	Yes	Yes, when integrity protection is used
NAT Compatibility	Limited	Better suited for NAT environments
Performance Overhead	Lower	Higher due to encryption
Main Purpose	Authentication and integrity	Confidentiality, integrity, and authentication

Security Services

AH mainly protects the packet from unauthorized modification and verifies the authenticity of the sender. However, the actual data remains visible.

ESP encrypts the packet payload, preventing unauthorized users from reading confidential information. It can also provide integrity and authentication.

Performance Overhead

AH has lower computational overhead because it does not perform encryption. ESP requires additional processing because the payload must be encrypted and decrypted.

However, modern systems can efficiently handle encryption, making the additional overhead acceptable for most enterprise networks.

Suitability for Different Network Scenarios

AH is suitable for:

- Communication where integrity is important.
- Environments where confidentiality is not required.
- Limited or specialized network scenarios.

ESP is suitable for:

- VPN communication.
- Secure branch-to-branch connections.
- Financial and business data.
- Communication over public networks.

Recommendation for VPN Deployment

For VPN deployment, **ESP is more appropriate** because VPN communication usually requires confidentiality in addition to integrity and authentication.

Branch Office A

|

| ESP Encryption

v

Public Internet

|

v

Branch Office B

Conclusion

ESP is generally the preferred protocol for VPN deployment because it provides stronger and more complete security services. AH may be used in situations where only authentication and integrity are required.

2. Compare transport mode and tunnel mode in IPSec architecture. Evaluate their effectiveness in securing host-to-host versus network-to-network communication.

IPSec supports two modes of operation:

1. Transport Mode
2. Tunnel Mode

Comparison

Feature	Transport Mode	Tunnel Mode
Protected Portion	Payload	Entire original IP packet
Original IP Header	Visible	Encapsulated and protected
Additional IP Header	No	Yes
Overhead	Lower	Higher
Main Use	Host-to-host	Network-to-network
VPN Suitability	Limited	Highly suitable
Internal Network Protection	Lower	Higher

Transport Mode

In transport mode, only the payload is protected.

Original Packet:

[IP Header][Payload]

After IPSec:

[IP Header][Protected Payload]

The original IP header remains visible so that routers can deliver the packet.

Suitable for

- Host-to-host communication.
- Communication between two individual servers.
- Internal secure communication.

Advantages

- Lower packet overhead.
- Better performance.
- Suitable for direct communication between hosts.

Limitation

The original source and destination information remains visible.

Tunnel Mode

In tunnel mode, the complete original IP packet is encapsulated inside a new packet.

Original Packet:

[Original IP Header][Payload]

Tunnel Mode:

[New P Header][Protected Original IP Header + Payload]

Suitable for

- Site-to-site VPN.
- Branch office communication.
- Network-to-network communication.

Advantages

- Protects the complete original packet.
- Hides internal IP addresses.
- Provides stronger privacy.
- Suitable for communication over the public internet.

Comparison of Effectiveness

For **host-to-host communication**, transport mode is effective because the communication occurs directly between two systems.

For **network-to-network communication**, tunnel mode is more effective because entire networks can communicate securely through VPN gateways.

Network A

|

[VPN Gateway]

|

| Tunnel Mode

|

Public Internet

|

[VPN Gateway]

|

Network B

3. Perform a comparative analysis of Pretty Good Privacy (PGP) and S/MIME in terms of key management, trust models, scalability, and enterprise usability. Recommend the most suitable approach for a corporate email system.

PGP and S/MIME are widely used technologies for securing email communication. Both provide encryption and digital signatures, but they differ in their trust models and key management.

Comparison

Feature	PGP	S/MIME
Full Form	Pretty Good Privacy	Secure/Multipurpose Internet Mail Extensions
Encryption	Yes	Yes
Digital Signatures	Yes	Yes
Key Management	User-managed	Certificate-based
Trust Model	Web of Trust	Certificate Authority
Scalability	Moderate	High
Enterprise Integration	Moderate	High
User Administration	More complex	Easier with centralized management
Best Use Case	Individuals and small groups	Enterprises and organizations

Key Management

PGP

PGP uses public and private key pairs. Users generally manage their own keys and verify the authenticity of other users.

User A

|

Trust Relationship

|

User B

|

Trust Relationship

|

User C

This approach is called the **Web of Trust**.

S/MIME

S/MIME uses digital certificates issued by a trusted Certificate Authority.

User

|

Digital Certificate

|

Certificate Authority

|

Trusted Root

This provides centralized identity verification.

Trust Models

PGP depends on users trusting and verifying each other's keys.

S/MIME depends on a hierarchy of trusted Certificate Authorities.

For a large organization, the centralized trust model of S/MIME is easier to manage.

Scalability

PGP may become difficult to manage when thousands of employees are involved because users may need to handle keys and trust relationships.

S/MIME is more scalable because certificates can be issued, renewed, and revoked centrally.

Enterprise Usability

S/MIME integrates well with corporate email systems and centralized identity infrastructure.

PGP provides flexibility but requires more user involvement.

Recommendation

For a corporate email system, S/MIME is the most suitable approach.

Justification

1. Centralized certificate management.
2. Better enterprise scalability.
3. Easier administration.
4. Certificate revocation support.
5. Integration with organizational email systems.
6. Strong identity verification using Certificate Authorities.

4. Compare rule-based and machine learning-based email spam detection techniques. Analyze their effectiveness in handling evolving spam patterns and minimizing false positives.

Spam detection systems can use traditional rule-based techniques or Machine Learning techniques.

Comparison

Feature	Rule-Based Detection	ML-Based Detection
Detection Method	Predefined rules	Learns from data
Adaptability	Low	High
New Spam Detection	Difficult	Better

Feature	Rule-Based Detection	ML-Based Detection
Complexity	Simple	Higher
Training Data	Not required	Required
Maintenance	Manual rule updates	Model retraining
False Positives	Can be high	Can be reduced with proper training
Handling Evolving Spam	Limited	More effective

Rule-Based Spam Detection

Rule-based systems use predefined conditions.

Example:

IF email contains "FREE"

AND email contains "WIN"

THEN classify as SPAM

Advantages

- Easy to implement.

- Fast processing.
- No training dataset required.
- Rules are easy to understand.

Disadvantages

- Attackers can bypass known rules.
- Requires frequent manual updates.
- New spam patterns may not be detected.
- Can generate false positives.

For example, a legitimate marketing email containing the word "free" could be incorrectly classified as spam.

Machine Learning-Based Spam Detection

Machine Learning models learn patterns from previously classified emails.

Possible features include:

- Number of suspicious links.
- Sender reputation.
- Spam keywords.
- Email header information.

- SPF, DKIM, and DMARC results.
- Attachment type.
- Number of capital letters.
- Frequency of suspicious words.

Possible algorithms include:

- Naive Bayes
- Logistic Regression
- Random Forest
- Support Vector Machine

Advantages

- Can identify complex patterns.
- Adapts better to new spam techniques.
- Can combine multiple features.
- Can improve over time through retraining.

Disadvantages

- Requires training data.

- Requires model maintenance.
- May be computationally more expensive.
- Poor training data can increase false positives.

Handling Evolving Spam

Rule-based systems struggle when attackers modify their messages.

For example:

FREE MONEY

can be changed to:

FR33 M0N3Y

A simple rule may fail to detect this variation.

Machine Learning models can analyze multiple patterns and features, making them more effective against evolving spam.

False Positive Handling

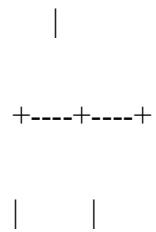
A good spam detection system should use multiple risk levels.

Incoming Email

|

v

Spam Score



Low Risk High Risk

Deliver Quarantine

Instead of immediately deleting suspicious emails, high-risk messages can be moved to quarantine for review.

Recommendation

A **hybrid approach** is the most effective:

Rule-Based Filtering

+

Machine Learning

+

Header Authentication

=

Advanced Spam Detection

5. Compare manual key management and automated key management (e.g., using IKE) in IPSec. Evaluate their impact on security, scalability, and operational complexity in large organizations.

IPSec requires cryptographic keys and Security Associations to establish secure communication. Key management can be performed manually or automatically using protocols such as IKE.

Comparison

Feature	Manual Key Management	Automated Key Management using IKE
Configuration	Manual	Automatic
Scalability	Low	High
Human Errors	High	Low
Key Renewal	Manual	Automatic
Security Association Creation	Manual	Automatic
Key Rotation	Difficult	Easy
Operational Complexity	High in large networks	Lower after setup

Feature	Manual Management	Key Automated Key Management using IKE
Enterprise Suitability	Small networks	Large organizations

Manual Key Management

In manual key management, administrators configure cryptographic keys directly on each device.

Process

Administrator

|

Configure Key

|

Device A

|

Configure Same Key

|

Device B

Advantages

- Simple for small networks.
- No additional key exchange protocol required.
- Can be useful in static and controlled environments.

Disadvantages

- High risk of configuration errors.
- Difficult to manage expired keys.
- Difficult to scale.
- Key rotation requires manual updates.
- A compromised key requires manual replacement on multiple devices.

Automated Key Management Using IKE

Internet Key Exchange automates authentication, Security Association creation, and key generation.

Process

Device A

|

| IKE Authentication

|

Device B

|

v

Secure Key Exchange

|

v

Create Security Association

|

v

Generate Session Keys

|

v

Secure IPSec Communication

Advantages

1. Automatic key generation.

2. Automatic Security Association management.
3. Automatic key renewal.
4. Reduced human error.
5. Better scalability.
6. Faster recovery when keys expire.
7. Suitable for large enterprise networks.

Security Impact

Manual key management increases the risk of:

- Weak keys.
- Reused keys.
- Configuration mistakes.
- Delayed key replacement.

Automated IKE-based management improves security by allowing regular key rotation and automatic negotiation of security parameters.

Scalability

Consider an organization with multiple branch offices.

With manual management:

Branch A ↔ Branch B

Branch A ↔ Branch C

Branch B ↔ Branch C

Administrators must configure multiple keys manually.

As the number of locations increases, management becomes increasingly complex.

With IKE, Security Associations and session keys can be negotiated automatically.

Operational Complexity

Manual management may initially appear simple, but operational complexity increases rapidly as the organization grows.

Automated management requires initial configuration but significantly reduces long-term administrative work.